

Empirical Analysis of Transformers on Graph Connectivity

Part VI: Learning to Reason — Which Data Teaches a Connectivity Path?

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Abstract

(Provisional.) The first five reports asked *what* algorithm the standard connectivity transformer ends up computing (a parallel, distance-bounded matrix powering, not a sequential traversal; Report V). This report turns the question around and asks *how it learns to reason*: which training data let an $L = 2$ model acquire a connectivity “path” it was never shown, and where instead the limit is simply not having seen the structure. We organise the study around two ideas the architecture makes concrete — **parallelism** (several routes between two endpoints) and **repeated/iterated structure** (a chain of locally-decided links) — and around one methodological commitment: every experiment trains on a single, clearly named distribution, never an opaque random mixture. This is the concluding report of the series, so we keep the experiment count small and each one pointed at a single claim.

1 Introduction

1.1 What Reports I–V established

We treat the following as established (each multi-seed, documented in Reports I–V); the new study must stay consistent with them.

- **The data lever does not help the standard transformer generalise**, and whether a run finds the generalising solution is **seed-dominated**; the diameter filter only speeds up convergence (Reports II–III).
- **There is a sharp, architectural distance wall at $\approx 3^L$ ($= 9$ at $L = 2$)**, which moves with depth in a causal sweep and appears even on families seen $\sim 10^8$ times in training — steering the data cannot move it (Reports III–IV).
- **Difficulty has two separable axes**: distance-vs-capacity and density-vs-training; the spectral gap is not a third axis on natural graphs (collinear with the diameter), though a purpose-built probe can isolate a bottleneck (Report IV).
- **A similarity read-out doubles the reach to $2 \cdot 3^L$** (a meet-in-the-middle of two 3^L -hop neighbourhoods) but cannot cross a bottleneck (Report IV).
- **The model computes distance-bounded matrix powering, not a bounded traversal** (DFS rejected; Report V). The bridged-clique “node budget” (~ 6 – 7) that looked like a second capacity is a **soft, learned shortcut**: it moves with the read-out, *dissolves once the structure is in training* (and the learned skill then transfers to an unseen bridge), and does not grow with depth.

1.2 The new question: how does it learn a reasoning path, and from what data?

Report V settled *which* computation the model runs. The two facts that make this report possible are the last one above: putting a structure in training can teach a *within-capacity*

propagation skill the model otherwise skips, and that skill can *transfer* to structures it never saw. So the limiting factor for a given “reasoning path” (propagate a connection along a particular shape) is often not the architecture but *which data the model was trained on*. This report asks that directly, through two lenses the transformer makes concrete:

- **Parallelism.** A transformer resolves every pair at once. Report IV’s confound-free probe already hinted that, beyond the capacity, *several* parallel routes between two endpoints make a connection the model could not confirm over a single route. If real and robust, this is a clean statement about *how* the model reasons: redundant evidence (more routes) is easier than a single thin thread, even at fixed distance. Report IV measured this on one seed and on data partly saturated to 1.0; here we redo it properly.
- **Iterated/repeated structure.** Report V showed the model can be taught the single-bridge “two cliques are one component iff the bridge is there” decision. A natural next step is whether that local decision *chains*: a sequence clique–bridge–clique–bridge–... asks the model to find, at each stage, the node that carries the link to the next block — a forced hand-off, repeated. This probes whether a learned local rule composes into a longer path of reasoning.

1.3 The claims this report argues

We keep the experiment set small and each one aimed at one claim. The intended spine:

1. **More parallel routes help** the model resolve a connection, *even when it never saw such multi-route graphs in training* (parallelism aids the reasoning); and there is a measurable **sample cost** to learning it when the graphs *are* shown.
2. **Truncated routes in training hurt**: mixing in dead-end paths (a route that does not reach the far endpoint) confuses the connection skill rather than helping it.
3. **Failing to handle the bridged cliques is “never seen”, not “too hard”**: like a too-dense ER graph it is an out-of-distribution gap, not an intrinsic difficulty — shown by the fact that once the structure is in training the model handles it (Report V), and we push this to repeated/varied versions.
4. **A learned single bridge composes**: a model that learns one bridged clique can handle it *repeated* (chains of bridged cliques) and with *different block types* (e.g. dense ER blocks instead of complete cliques), as long as the construction stays within the distance capacity (≤ 9). (*To be tested — this is a hypothesis, not yet a result.*)

2 Setup and the data principle

Model. Unless noted we use the same base model as Reports III–V: the RoBERTa-faithful standard transformer with the linear read-out, $L = 2$, single head, $d_{\text{model}} = 512$, normalised-ReLU attention, trained online at $n = 20$ (and $n = 40$ where the canvas needs to be larger) for 10^6 steps. The input is the self-loop-augmented adjacency matrix A ; the target is the connectivity matrix R . Metrics are the usual exact-match, pairwise, and per-distance/per-pair accuracies, plus, where relevant, the accuracy on a *single designated pair* (the two endpoints of a multi-route graph).

The data principle (new in this report). Every experiment trains on a *single, explicitly named* graph distribution — pure ER, cliques, bridged cliques, the multi-route “multipath” family, or a clearly stated and motivated combination of a few of them. We do *not* use the opaque uniform-over-nine-families “mixed” stream of Reports III–V as a default training set:

a result is only worth reporting if the training distribution is simple enough to state in one sentence and motivate. Where an earlier report’s number came from the mixed stream and we reuse it, we say so and treat it as a baseline, not as the clean training condition.

The two core families.

- **Multipath** (the parallel family, from Report IV): two terminals a, b joined by k internally-disjoint paths each of length ℓ , so the distance $\text{dist}(a, b) = \ell$ is fixed and only the number of routes k varies (the effective resistance is ℓ/k). The terminals are padded to a fixed degree and the rest of the canvas is filled sparsely, so k is the only thing that changes. A **truncated** multipath replaces some routes with dead ends that do not reach b .
- **Bridged cliques** (the iterated family, from Report V): two cliques (or two internally-dense blocks) of size c joined by a single bridge edge — one component. We extend it two ways here: a **chain** of cliques joined by successive bridges (clique–bridge–clique–...), and **thicker bridges** (more than one edge), keeping every cross distance ≤ 9 so the target is always within the matrix-power capacity.

3 Plan of experiments

The experiments fall into three threads, plus a short exploratory one.

Thread A — Parallelism: the multipath family. Redo Report IV’s parallel-route probe properly (multi-seed, and only on the route/length cells that are actually informative), then ask how many multipath graphs it takes to *learn* the family, and whether truncated routes in training hurt.

Thread B — Asymmetric two-chains. Explain a specific puzzle from Report IV: a graph split into two paths of very unequal length (e.g. 4 and 36) appeared *easier* than a more balanced split (e.g. 17 and 23). We analyse the existing data and run a controlled experiment, reading it as a statement about how the model reasons along a path.

Thread C — Iterated structure: chained and thickened bridged cliques. Train on a clean single-bridge family and test on repeated bridges (chains of cliques) and on thicker bridges and different block types (dense ER blocks), to see whether a learned local link composes.

Thread D — Trees (exploratory). A short look at tree-structured graphs as another reasoning-path shape.

4 Results

(All subsections below are stubs: the runs are planned, not yet collected.)

4.1 Thread A.1 — Parallel routes help, even unseen in training

Question. At a fixed endpoint distance, does adding parallel routes between a and b make the connection easier to resolve — and does it, even for a model that never saw a multi-route graph in training (pure ER)?

[To run — data pending. Report IV’s single-seed version found that, beyond the capacity, one route fails (≈ 0.55) while two or three succeed; this redo establishes whether that holds across seeds and identifies which route/length cells are informative rather than saturated.]

4.2 Thread A.2 — The sample cost of learning the multipath connectivity

Question. When multipath graphs *are* in training, how many does it take to learn the connectivity exactly — for the single (a, b) pair, and for the whole matrix?

[To run — data pending.]

4.3 Thread A.3 — Truncated routes in training confuse the model

Question. If some training routes are dead ends (truncated), does that degrade what the model learns about the genuine routes?

[To run — data pending.]

4.4 Thread B — Why an unbalanced two-chain split is easier

Question. Report IV suggested that splitting n nodes into two paths of very unequal length is *easier* than a balanced split. Why, and what does it say about how the model reasons along a path?

[To analyse the existing data, then run — pending.]

4.5 Thread C.1 — Chained bridged cliques: does a learned bridge compose?

Question. A single bridged clique is a learnable “iff” decision (Report V). Does it *chain* — can the model carry the link across clique–bridge–clique–..., finding at each stage the node that hands off to the next block?

[To run — data pending.]

4.6 Thread C.2 — Thicker bridges and different blocks

Question. Does a thicker bridge (more than one edge) make the hand-off easier, and do dense ER blocks behave like complete cliques — all kept within the distance capacity?

[To run — data pending.]

4.7 Thread D — Trees (exploratory)

Question. How does the model handle tree-structured connectivity, as a third reasoning-path shape beside parallel routes and chained bridges?

[Exploratory — scope and runs pending.]

5 Verdict (to write last)

[To write last, once Threads A–C are in.]